

MGT201 Financial Management

Formulas Lecture 1 to 22

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1. Fundamental Accounting Equation and Double Entry Principle.

- $\text{Assets} + \text{Expense} = \text{Liabilities} + \text{Shareholders' Equity} + \text{Revenue}$

$\text{Liabilities} = \text{Equity} = \text{Net Worth}$

$\text{Revenue} - \text{Expense} = \text{Income}$

1. Statement of Retained Earnings or Shareholders' Equity Statement

$\text{Total Equity} = \text{Common Par Stock Issued} + \text{Paid In Capital} + \text{Retained Earnings}$

2. Current Ratio:

$= \text{Current Assets} / \text{Current Liabilities}$

3. Quick/Acid Test ratio:

$= (\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$

4. Average Collection Period:

$= \text{Average Accounts Receivable} / (\text{Annual Sales} / 360)$

5. PROFITABILITY RATIOS:

Profit Margin (on sales):

$= [\text{Net Income} / \text{Sales}] \times 100$

Return on Assets:

$= [\text{Net Income} / \text{Total Assets}] \times 100$

Return on equity:

$= [\text{Net Income} / \text{Common Equity}]$

6. ASSET MANAGEMENT RATIOS

Inventory Turnover:

= Sales / inventories

Total Assets Turnover:

= Sales / Total Assets

7. DEBT (OR CAPITAL STRUCTURE) RATIOS:

Debt-Assets:

= Total Debt / Total Assets

Debt-Equity:

= Total Debt / Total Equity

Times-Interest-Earned:

= EBIT / Interest Charges

8. Market Value Ratios:

Price Earning Ratio:

= Market Price per share / *Earnings per share

Market /Book Ratio:

= Market Price per share / Book Value per share

*Earning Per Share (EPS):

= Net Income / Average Number of Common Shares Outstanding

9. M.V.A (Market Value Added):

MVA (Rupees) = Market Value of Equity – Book Value of Equity Capital

10. E.V.A (Economic Value Added):

EVA (Rupees) = EBIT (or Operating Profit) – Cost of Total Capital

11. Interest Theory:

• Economic Theory:

$$i = iRF + g + DR + MR + LP + SR$$

– i is the nominal interest rate generally quoted in papers. The “real” interest rate = $i - g$

Here i = market interest rate

g = rate of inflation

DR = Default risk premium

MR = Maturity risk premium

LP = Liquidity preference

SR = Sovereign Risk

The explanation of these determinants of interest rates is given as under:

12. Market Segmentation:

- **Simple Interest (or Straight Line):**

$$FV = PV + (PV \times i \times n)$$

- **Discrete Compound Interest:**

Annual (yearly) compounding:

$$FV = PV \times (1 + i)^n$$

Monthly compounding:

$$FV = PV \times (1 + (i / m))^{m \times n}$$

- **Continuous (or Exponential) Compound Interest:**

- $FV \text{ (Continuous compounding)} = PV \times e^{i \times n}$

13. Estimated current assets for the next year

$$= [\text{Current assets for the current year} / \text{Current sales}] \times \text{Estimated sales for the next year}$$

14. Expected Estimated retained earnings

$$= \text{estimated sales} \times \text{profit margin} \times \text{plowback ratio}$$

15. Estimated discretionary financing

$$= \text{estimated total assets} - \text{estimated total liabilities} - \text{estimated total equity}$$

16. G (Desired Growth Rate)

$$= \text{return on equity} \times (1 - \text{pay out ratio})$$

17. CASH FLOW STATEMENT

Net Income

Add Depreciation Expense

Subtract Increase in Current Assets:

Increase in Cash

Increase in Inventory

Add Increase in Current Liabilities:

Increase in A/c Payable

Cash Flow from Operations

Cash Flow from Investments

Cash Flow from Financing

Net Cash Flow from All Activities

18. Interest Rates for Discounting Calculations

- Nominal (or APR) Interest Rate = i_{nom}

- Periodic Interest Rate = i_{per}

It is defined as

$$i_{\text{per}} = (i_{\text{nominal Interest rate}}) / m$$

Where

m = no. of times compounding takes place in 1 year i.e.

If semi-annual compounding then $m = 2$

- Effective Interest Rate = i_{eff}

$$i_{\text{eff}} = [1 + (i_{\text{nom}} / m)]^m - 1$$

19. Calculating the NPV of the Café Business for 1st Year:

NPV = Net Present Value (taking Investment outflows into account)

NPV = -Initial Investment + Sum of Net Cash Flows from Each Future Year.

$$\text{NPV} = -I_0 + \text{PV}(\text{CF1}) + \text{PV}(\text{CF2}) + \text{PV}(\text{CF3}) + \text{PV}(\text{CF4}) + \dots + \infty$$

20. Annual Compounding (at end of every year):

$$FV = CCF (1 + i)^n - 1$$

Annual Compounding (at end of every year)

$$PV = FV / (1 + i)^n \cdot n = \text{life of Annuity in number of years}$$

21. Multiple Compounding:

$$\text{Future Value of annuity} = CCF (\text{constant cash flow}) * (1 + (i/m)^{m*n} - 1/i/n$$

Multiple Compounding:

$$PV = FV / [1 + (i/m)]^{m*n}$$

22. Future value of perpetuity:

$$= \text{constant cash flow} / \text{interest rate}$$

23. Future value by using annuity formula

$$FV = CCF [(1+i)^n - 1] / i$$

24. Return on Investments:

$$ROI = (\Sigma CF/n) / I_o$$

25. Net Present Value (NPV):

$$NPV = -I_o + \Sigma CF_t / (1+i)^t$$

Detail

$$NPV = -I_o + CF_t / (1+i)^t = -I_o + CF_1 / (1+i) + CF_2 / (1+i)^2 + CF_3 / (1+i)^3 + \dots$$

-I_o = Initial cash outflow

i = discount / interest rate

t = year in which the cash flow takes place

26. Probability Index:

$$PI = [\Sigma CF_t / (1+i)^t] / I_o$$

27. Internal Rate Of Return (IRR) Equation:

$$NPV = -I_o + CF_1 / (1+IRR) + CF_2 / (1+IRR)^2$$

28. Internal Rate of Return or IRR:

The formula is similar to NPV

$$NPV = 0 = -I_0 + CF_t / (1+IRR)_t = -I_0 + CF_1/(1+IRR) + CF_2/(1+IRR)^2 + \dots$$

29. Modified IRR (MIRR):

$$(1+MIRR)^n = \frac{\text{Future Value of All Cash Inflows} \dots}{\text{Present Value of All Cash Outflows}}$$

$$(1+MIRR)^n = \frac{CF_{in} * (1+k)^{n-t}}{CF_{out} / (1+k)^t}$$

30. Equivalent Annual Annuity Approach:

$$EAA \text{ FACTOR} = (1+i)^n / [(1+i)^n - 1]$$

Where n = life of project & i = discount rate

BONDS' VALUATION

The relationship between present value and net present value

$$31. NPV = -I_0 + PV$$

32. Present Value formula for the bond:

$$PV = \sum_{t=1}^n CF_t / (1+rd)^t = CF_1/(1+rd) + CF_n/(1+rd)^2 + \dots + CF_n / (1+rd)^n + PAR / (1+rd)^n$$

In this formula

PV = Intrinsic Value of Bond or Fair Price (in rupees) paid to invest in the bond. It is the Expected or Theoretical Price and NOT the actual Market Price.

rd = it is very important term which you should understand it care fully. It is Bondholder's (or Investor's) Required Rate of Return for investing in Bond (Debt). As conservative you can choose minimum interest rate. It is derived from Macroeconomic or Market Interest Rate. Different from the Coupon Rate!

Recall Macroeconomic or Market Interest Theory: $i = i_{RF} + g + DR + MR + LP + SR$

CF = cash flow = Coupon Receipt Value (in Rupees) = Coupon Interest Rate x Par Value.

Represents cash receipts (or in-flow) for Bondholder (Investor). Often times an ANNUITY

pattern Coupon Rate derived from Macroeconomic or Market Interest Rate. The Future Cash Flows from a bond are simply the regular Coupon Receipt cash in-flows over the life of the Bond. But, at Maturity Date there are 2 Cash In-flows: (1) the Coupon Receipt and (2) the Recovered Par or Face Value (or Principal)

n = Maturity or Life of Bond (in years)

In the next lectures, you would study that how the required rate of return is related to market rate of return

33. Calculate the PV of Coupons from the FV Formula for Annuities (with multiple compounding within 1 year):

$$FV = CCF (1 + rD/m)^{nxm} - 1/rD/m$$

Use Monthly Basis for this example. $n = 1$ year $m = 12$ months

CCF = Constant Cash Flow = Rs 1,000 = Monthly Coupon

rD = Annual Nominal Required Rate of Return for investment in Bond (Debt) = 10% pa.

Periodic Monthly Required Rate of Return is $rD/m = 10/12 = 0.833\% = 0.00833$ p.m.

$m = 12$ months

34. YTM = Total or Overall Yield:

= Interest Yield + Capital Gains Yield

35. Interest Yield or Current Yield:

= Coupon / Market Price

36. Capital Gains Yield:

= YTM - Interest Yield

$$37. FV = CCF[(1 + rD/m)^n - 1]/rD/m$$

$N = 1$ year, $m =$ no. of intervals in a year = 12

CCF = constant cash flow

n = Maturity or Life of Bond (in years)

38. Call:

=par value +I, year coupon receipts

Another thing to keep in mind is that YTM has two components first is

39. YTM:

=interest yield on bond +capital gain yield on bond

1. INTEREST YIELD =annual coupon interest /market price

2. CAPITAL YIELD =YTM –INTEREST YIELD

40. Perpetual Investment in Preferred Stock

– $PV = DIV_1 / r_{PE}$

41. Perpetual Investment in Common Stock:

$PV = DIV_1/(1+r_{CE}) + DIV_2/(1+r_{CE})^2 + \dots + DIV_n/(1+r_{CE})^n + P_n/(1+r_{CE})^n$

$PV = P_0^* =$ Expected or Fair Price = Present Value of Share, DIV_1 = Forecasted Future Dividend at end of Year 1, DIV_2 = Expected Future Dividend at end of Year 2, ..., P_n = Expected Future Selling Price, r_{CE} = Minimum Required Rate of Return for Investment in the Common Stock for you (the investor). Note that Dividends are uncertain and $n = \text{infinity}$

42. PV (Share Price) = Dividend Value + Capital Gain.

Dividend Value is derived from Dividend Cash Stream and Capital Gain / Loss from Difference between Buying & Selling Price.

43. Simplified Formula (P_n term removed from the equation for large investment durations

i.e. $n = \text{infinity}$):

$PV = DIV_1/(1+r_E) + DIV_2/(1+r_E)^2 + \dots + DIV_n/(1+r_E)^n$

= $DIV_t / (1 + r_E)^t$, t = year. Sum from $t=1$ to n

44. Fair Value. Dynamic Equilibrium.

If **Market Price > Fair Value** then Stock is Over Valued

Share Price Valuation -Perpetual Investment in Common Stock:

45. Zero Growth Dividends Model:

$$\text{DIV1} = \text{DIV 2} = \text{DIV3}$$

46. The Formula for common stock

$$\begin{aligned}\text{PV} &= \text{Po}^* = \text{DIV1} / (1 + r\text{CE}) + \text{DIV1} / (1 + r\text{CE})^2 + \text{DIV1} / (1 + r\text{CE})^3 + \dots + \dots \\ &= \text{DIV 1} / r\text{CE}\end{aligned}$$

47. Dividends Cash Flow Stream grows according to the Discrete Compound Growth

Formula

$$\text{DIV}_{t+1} = \text{DIV}_t \times (1 + g)^t$$

t = time in years.

48. Zero Growth Model Pricing

$$\text{PV} = \text{Po}^* = \text{DIV1} / r\text{CE}$$

49. Constant Growth Model Pricing

$$\text{PV} = \text{Po}^* = \text{DIV1} / (r\text{CE} - g)$$

50. Dividends Pricing Models:

Zero Growth: $\text{Po}^* = \text{DIV1} / r\text{CE}$ (Po^* is being estimated)

51. $r\text{CE}^* = \text{DIV1} / \text{Po}$ ($r\text{CE}^*$ is being estimated)

Similarly,

52. Constant Growth: $\text{Po}^* = \text{DIV1} / (r\text{CE} - g)$

$$r\text{CE}^* = (\text{DIV 1} / \text{Po}) + g$$

use this formula to calculate the required rate of return.

53. Gordon's Formula:

$$r\text{CE}^* = (\text{DIV 1} / \text{Po}) + g$$

In this the first part

($\text{DIV 1} / \text{Po}$) is the dividend yield

g is the Capital gain yield.

54. Earning per Share (EPS) Approach:

$$PV = P_0^* = EPS_1 / r_{CE} + PVGO$$

P_0 = Estimated Present Fair Price,

EPS_1 = Forecasted Earnings per Share in the next year (i.e. Year 1),

r_{CE} = Required Rate of Return on Investment in Common Stock Equity.

PVGO = Present Value of Growth Opportunities. It means the Present Value of Potential

Growth in Business from Reinvestments in New Positive NPV Projects and Investments PVGO is perpetuity formula.

The formula is

$$PVGO = NPV_1 / (r_{CE} - g) = [-I_0 + (C/r_{CE})] / (r_{CE} - g)$$

In this PVGO Model: Constant Growth “g”. It is the growth in NPV of new Reinvestment Projects (or Investment). $g = \text{plowback} \times ROE$

Perpetual Net Cash Flows (C) from each Project (or reinvestment).

I_0 = Value of Reinvestment (Not paid to share holders)

$$= P_b \times EPS$$

Where P_b = Plough back = 1 – Payout ratio

Payout ratio = (DIV/EPS) and

$$55. EPS \text{ Earnings per Share} = (NI - DIV) / \# \text{ Shares of Common Stock Outstanding}$$

Where NI = Net Income from P/L Statement and DIV = Dividend, $RE_1 = RE_0 + NI_1 + DIV_1$

ROE = Net income / # Shares of Common Stock Outstanding.

$$56. NPV_1 = [-I_0 + (C/r_{CE})] / (r_{CE} - g)$$

If we compare it with the traditional NPV formula

$-I_0$ = Value of initial investment

(C/r_{CE}) = present value formula for perpetuities where you assume that you are generating the net cash

inflow of C every year.

$$C = \text{Forecasted Net Cash Inflow from Reinvestment} = I_0 \times ROE$$

Where ROE = Return on Equity = NI / Book Equity of Common Stock Outstanding

57. Range of Possible Outcomes, Expected Return:

Overall Return on Stock = Dividend Yield + Capital Gains Yield (Gordon's Formula)

58. Expected ROR = $\langle r \rangle = \sum p_i r_i$

Where p_i represents the Probability of Outcome "i" taking place and r_i represents the Rate of Return (ROR) if Outcome "i" takes place. The Probability gives weight age to the return. The Expected or Most Likely ROR is the SUM of the weighted returns for ALL possible Outcomes.

59. Stand Alone Risk of Single Stock Investment:

Risk = Std Dev = $\sqrt{\sum (r_i - \langle r \rangle)^2 p_i}$. = Summed over each possible outcome "i" with return " r_i " and probability of occurrence " p_i ." $\langle r \rangle$ is the Expected (or weighted average) Return

60. Possible Outcomes Example Continued:

Measuring Stand Alone Risk for Single Stock Investment

Std Dev = $\sigma = \sqrt{\sum (r_i - \langle r \rangle)^2 p_i}$.

61. Coefficient of Variation (CV):

= Standard Deviation / Expected Return.

$CV = \sigma / \langle r \rangle$.

$\langle r \rangle = \text{Exp or Weighted Avg ROR} = \sum p_i r_i$

62. Risk Basics

Risk = Std Dev = $\sigma = \sqrt{\sum (r_i - \langle r \rangle)^2 p_i}$. = "Sigma"

63. Types of Risks for a Stock:

Types of Stock-related Risks which cause Uncertainty in future possible Returns & Cash Flows:

Total Stock Risk = Diversifiable Risk + Market Risk

64. Portfolio Rate of Return

Portfolio Expected ROR Formula:

$$r_P^* = r_1 x_1 + r_2 x_2 + r_3 x_3 + \dots + r_n x_n$$

65. Stock (Investment) Portfolio Risk Formula:

$$\sigma_P = \sqrt{X_A^2 \sigma_A^2 + X_B^2 \sigma_B^2 + 2(X_A X_B \sigma_A \sigma_B \rho_{AB})}$$

66. Efficient Portfolios:

$$r_P^* = x_A r_A + x_B r_B + x_C r_C$$

67.3-Stock Portfolio Risk Formula

3x3 Matrix Approach

Now, if we add another stock in the portfolio we can take a look

3-Stock Portfolio Risk Formula 3x3 Matrix Approach

	Stock A	Stock B	Stock C
Stock A	$X_A^2 \sigma_A^2$	$X_A X_B \sigma_A \sigma_B \rho_{AB}$	$X_A X_C \sigma_A \sigma_C \rho_{AC}$
Stock B	$X_B X_A \sigma_B \sigma_A \rho_{BA}$	$X_B^2 \sigma_B^2$	$X_B X_C \sigma_B \sigma_C \rho_{BC}$
Stock C	$X_C X_A \sigma_C \sigma_A \rho_{CA}$	$X_C X_B \sigma_C \sigma_B \rho_{CB}$	$X_C^2 \sigma_C^2$

To compute the **Portfolio Variance** for a 3-Stock Portfolio, just add up all the terms in every box. To compute the **Portfolio Risk** (Standard Deviation), simply take the Square Root of the Variance. **You can extend this Matrix Approach to calculate the Risk for a Portfolio consisting of any number of stocks.**

Terms in Boxes on Diagonal (Top Left to Bottom Right) are called “**VARIANCE**” terms associated with individual magnitude of risk for each stock.

Terms in all other (or NON-DIAGONAL) Boxes are called “**COVARIANCE**” terms which account for affect of one stock’s movement on another stock’s movement.